

# Test 10: Cross-validation stability

Verdict: PASS | Generated: 2026-09-02 13:50:03 | Suite: AB-Cloud v19 (Julia 1.12.0)

## What this test verifies

Statistical stability of the fit under data splits.

METHOD: Statistical stability of the fit under data splits.

Result Test 10 [HARDCORE pass 2]: 4 sub-checks, 0 failed, b(50000)=1.2126 ? PASS

Plots `plots/plot\_01` (.png 600dpi / .svg / .pdf)

- `plots/animation.gif` ? animated reveal of every plot of this test

Reproduce with: julia ab\_cloud\_v19.jl --test 10 --no-two-pass

## Independent verification guide

Every computation is logged in logs/computation\_log.txt (timestamps + deltas); raw console output in logs/stdout\_capture.txt.

(no log\_comp entries)

## FULL COMPUTATION LOG

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### CONSOLE CAPTURE

TEST 10 (secondary, HARDCORE): deep cross-validation audit

All sub-samples use index-aware Gram points (offset discipline)

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V1 deciles : 0 local increases over 10 segment means (limit 0)

V2 f-sweep : dev(0.3)=0.0019 ? dev(0.9)=0.0006 (must decrease)

devs: 0.0019 / 0.0011 / 0.0005 / 0.0006

V3 tripwire : correct-offset tail b=1.1095, wrong-offset b=19559.2539 ? +1762824.6% worse (limit ? 5%)

V4 even-odd : b\_even=1.208457 vs full 1.212553, dev=0.34% (limit 10%)

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V1 deciles PASS ? 0 increases / 10 segments

V2 f-convergence PASS ? dev 0.19% ? 0.06%

V3 offset tripwire PASS ? wrong offset +1762824.6% worse

V4 even-odd PASS ? dev = 0.34%

Test 10 [HARDCORE pass 2]: 4 sub-checks, 0 failed, b(50000)=1.2126 ? PASS